

Multi-Agent Debate CDR

每个agent要怎么设计, 是否agent都是必要的, 轻量级分类器是不是也行?

1.先根据popularity召回一个伪交互集合(效果好)

2.对于集合中的每一个伪交互, 利用不同agent进行分析, 输出prediction, reasoning, confidence

注: 输出要json格式, 防止prediction reasoning混在一起. $\text{confidence} = 1 - H(p)/\log 3$

$p = (0.6, 0.3, 0.1)$

其中:

- $\hat{y} \in \{\text{Strong}, \text{Weak}, \text{Reject}\}$: 三分类 prediction
- $c = 1 - H(p)/\log 3$: entropy-based confidence
- r : natural-language rationale

这样 **prediction** 是结构化监督, **confidence** 是可靠度, **reasoning** 是可解释性信息

温度系数要每个agent动态调整, 重叠用户验证集选择

prediction是仅限定A/B/C三种 (是否要微调 待定?) 不微调否则参数量太大

agent种类如下:

而是每个 agent 有自己的 observation:

A_sem : semantic evidence specialist

A_col : collaborative evidence specialist

A_ov : overlap-transfer evidence specialist

A_pop : popularity-bias auditor

尤其第四个我不建议叫 popularity expert。

它应该是:

Bias Auditor

专门判断:

这个 pseudo interaction 被召回, 到底是个性化证据充分, 还是只是因为 target item 太热门?

3.建立agent之间的关系

```
if prediction_a == prediction_b:
    Agreement

else if c_a > 0.7 and c_b > 0.7:
    Conflict

else if c_a - c_b > 0.3:
    Guidance: a → b

else if c_b - c_a > 0.3:
    Guidance: b → a

else:
    Weak Conflict / No Edge
```

权重计算如下:

```
w_agree = sqrt(c_a c_b) × (1 - JS(p_a,p_b))
```

```
w_conflict = sqrt(c_a c_b) × JS(p_a,p_b)
```

```
w_guide = (c_a-c_b) × JS(p_a,p_b)
```

4.异构图卷积 基于重叠用户 (能训练明白吗)

PCA降维 reasoning embedding取前64维 然后self-attention

或者设置judge agent微调(SFT/记忆)

5. $p = p_1 + \lambda p_2$ 设置阈值 不一定都要5个

6.最终加权进入到模型训练中

而是向 Overlap Agent 输入：

Semantic agent supports this interaction because XXX.
Collaborative agent supports it because XXX.
Bias auditor rejects it because XXX.
There is a high-confidence conflict. Re-evaluate the interaction based on your role.

于是：

state_t

→ relation construction

→ **new observation**

→ state_{t+1}



四个证据 Agent 的构建